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Methods Corner

Incorporating co-design in cardiovascular research: tips, tricks, and tools

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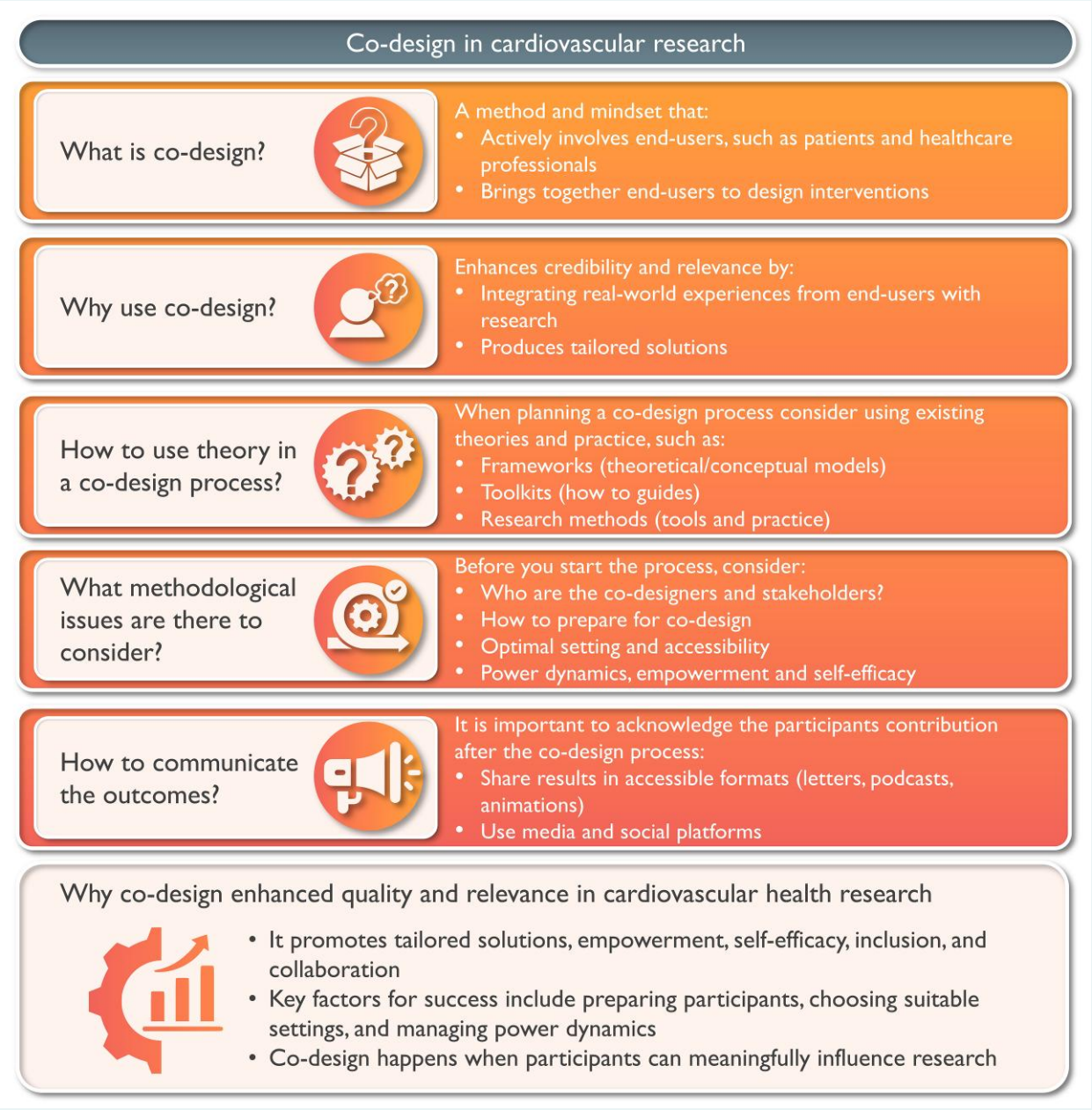
Co-design is increasingly used in cardiovascular health research. Conducting co-design is an iterative process that involves collaboration with end-users, such as clinicians and people with lived experience. However, co-design is logistically, practically, and methodologically challenging. This methods paper aims to clarify what co-design is, why and when to use co-design, and how co-design research can be conducted. This paper, therefore, provides practical tools and examples to deal with the challenges and achieve a constructive co-design process in cardiovascular research.

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Graphical Abstract



Keywords Cardiovascular health research • Co-design • Co-participation • Research method

Learning objectives

- Define co-design.
- Describe the rationale and appropriate contexts for using co-design.
- Provide insights and recommendations to researchers based on experience and examples.

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What is co-design?

Co-design is a method and a mindset that is increasingly used in health and medical research, including the field of cardiovascular disease.[1–4] In the context of research, co-design refers to the meaningful involvement of end-users at any stage of the research process, from planning to dissemination of findings.[5] This approach aims to enhance the relevance and impact of research by incorporating the perspectives and expertise of those directly affected by the research outcomes.[6] Instead of a clear-cut definition of the concept, co-design can be considered as the 'core ideas' involved when engaging in a co-design project. One such core idea is the active and meaningful involvement in a co-design process of end-users and other stakeholders in designing research, innovations, interventions, or health systems.[7] End-users can be patients, informal carers, people with lived experiences, or healthcare professionals and are those who will ultimately be affected by the result of the co-design.[6–8] Other stakeholders involved in collaborative activities could include those who can contribute to the design or development of the innovation or those who could influence the potential for implementation and translation. Other stakeholders involved in collaborative activities could include those who can contribute to the design or development of the research, innovations, interventions, or health systems or those who could influence the potential for implementation and translation. The intent with the involvement and collaboration is both methodological, by seeking to enhance the relevance and impact of research, and moral/ethical, by respecting the rights, voices, and lived experiences of those affected by the research.[6] A second core idea in co-design is the avoidance of involving end-users in a tokenistic manner, ensuring that their input is authentic and valued equally to that of other stakeholders. This includes recognizing end-users as 'experts by experience' and professionals or researchers as 'experts by training'.[9,10]

In addition to core ideas, core features of co-design are important to consider. One such core feature is the need for creativity, as co-design involves design processes aimed at generating new innovations—such as a new care pathway, models of care, or digital solutions for support—which inevitably require being creative. A second core feature is the need to be flexible to accommodate the process, as these are often iterative, requiring going back and forth between developing and testing ideas.[11] In a healthcare context, it can also be challenging for participants to engage in co-design processes according to a predetermined schedule, as they may be affected by chronic and acute health issues that require them to adjust their participation during the project, and some individuals may even need to withdraw from the project before its completion. A third core feature in co-design is the use of prototypes of interventions for testing ideas and for receiving feedback on possible changes to interventions.[12] Having something tangible may facilitate feedback and generate new ideas.

From our experience, it is important to emphasize these core ideas and core features so that the engagement of, and collaboration with, end-users and other stakeholders in research leads to genuine influence and impact on the outcomes of research projects. These core aspects are elaborated on in coming sections. Moreover, as healthcare settings may present challenges, as we sometimes include individuals who could be considered vulnerable and may to be able to contribute in the intended way, we also expand on this. Additionally, we focus on how power dynamics affect whose input carries the most weight, which means that we may need to be creative in how we genuinely include and collaborate to avoid tokenism.[13,14]

Why use co-design?

Co-designing research, innovations, interventions, or health systems offers a wide range of benefits, many of which stem from actively involving users in decision-making processes. Benefits will vary depending on the specific goals of the co-design initiative and the reasons for choosing this approach to user involvement. Firstly, end-users have valuable insights, experiences, and preferences that can greatly augment the quality of what is being designed.[11] More specifically, co-design has the potential to lead to 'tailored solutions' that enhance the possibility for interventions to meet the unique needs of the target population. If the co-design process also includes healthcare professionals, this could further ensure that the intervention is evidence based, which bolsters its value. Secondly, including end-users in research within public domains can be viewed as a 'civil right', as they are utilizing the solutions and innovations being developed.[15,16] Thirdly, end-users included in a co-design process can experience feelings of 'empowerment' and an increased sense of 'self-efficacy',[17] as well as becoming more knowledgeable and skilled, which adds important value within this context.[6] Fourthly, from the researcher's perspective, being involved in co-design can provide an enhanced understanding of the relevant insights of end-users, which in turn may add credibility to the research, enhancing overall trustworthiness and impact.[18]

Beyond these moral and methodological arguments, there is the question of whether co-design leads to better interventions and policies in terms of reach, effectiveness, implementation, translation, sustainability, and scalability. There is some evidence that the research process can be improved using co-design.[6] However, systematic and umbrella reviews on the topic do not provide a clear picture of the link between co-design and more effective or more sustainable interventions. Increasingly, there are efforts to create frameworks to evaluate the impacts of co-design; however, it is too early to assess their use[5] largely because of the lack of a standardized definition of co-design and the lack of robust evaluations.[6,19] The ambiguity regarding the impact of co-design may also be due to an absence of standardization and rigour in methods and evaluations of co-design and the impacts it has on intervention effectiveness.[20] In a scoping review that focused on the use of Experience Based Co-Design (EBCD), one of the more structured co-design methods, the authors found evidence for positive impacts of co-design. These included the production of improved materials/information, services, processes, and environments. They also reported improved intergroup dynamics and subjective wellness.[21] Recently, there have been increasing efforts to create frameworks for monitoring and evaluating the impact of co-design methods on the effectiveness and implementation of interventions and policy.[19] However, it is too early to evaluate the uptake of EBCD and other frameworks to conduct robust evaluations of the impacts of co-design.

Conceptual framework, toolkit, and method format

There are numerous methodologies for conducting co-design research. A summary of recommended conceptual frameworks, toolkits, and method formats is provided for further reading in *Table 1*.

Data can be collected during the co-design process by one person taking notes or by recording the meetings and then analysed with qualitative method such as qualitative content analysis depending on the study protocol.

Methodological considerations when using co-design

Who are the co-designers?

Co-designers generally encompass end-users and a varied group of other stakeholders. End-users can be patients, informal carers, people with lived experiences, or healthcare providers and are those affected by what is being designed. Typical stakeholders are (i) organizational

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Table 1 Conceptual frameworks, toolkits, and method formats recommended for a co-design process

Conceptual frameworks:

Community-Based Participatory Action Research[22]—a methodological and epistemological approach to applied community projects in which researchers and community members collaborate as equals in the research process.

- Patient-centred outcome research—research investigating the outcomes that are important to patients, with the rationale that clinical research is of higher quality when it is informed by perspectives of the end-users.[23]
- Participatory action research (PAR)—combines participation and action to understand and address societal issues. Emphasizes democratic processes in participation *with* others rather than research for research's sake conducted *on* people/communities.[24]
- The Experience-Based Co-Design (EBCD)[25,26] is a participatory research process that brings together healthcare staff and patients to collaboratively improve the quality of care in a six-stage process. This process includes gathering and analyzing patient narratives and identifying key areas for improvement.

Toolkits:

- The co-design guide and toolkit of Boyd et al.[27] is a five-step co-design process that involves end-users and stakeholders to understand their experiences, planning clear goals for the project, exploring detailed insights through various methods, developing creative solutions collaboratively, and deciding on the best interventions to implement.
- The Consumer Involvement and Engagement Toolkit.[28] This toolkit supports meaningful consumer involvement in clinical trials through a structured, stage-based process. It features an interactive map, practical tools, and templates for planning, delivering, and evaluating engagement. Resources are tailored for both researchers and consumers to promote collaboration.

Method formats:

- World café method.[29] This is a simple, effective, and flexible process for facilitating dialogue with larger groups to gain opinion and works flexible and effective for facilitating large group dialogue, based on seven design principles. Each element of the method has a specific purpose and corresponds to one or more of the design principles. Desk-based reviews: a review of relevant literature relating to a policy, intervention, or condition that can be conducted early in the project as part of an exploratory situational analysis.
- Community mapping: a useful approach to mapping relationships, assets, and spaces that usually involves a form of geographic mapping.[30]
- Workshops: this is often considered as a 'classic' co-design format. A group of end-users and/or other stakeholders and researchers are brought together to design, develop, or adapt ideas, policies, or interventions. Workshops can take many forms and have many names such as idea groups, discussion groups, or qualitative focus groups, and there are a wide range of methods for conducting co-design workshops.
- Interviews: Individual qualitative interviews with end-users and stakeholders can be useful during all phases of a co-design project but particularly in the exploratory or evaluation stages.

stakeholders, those who will be responsible for delivering a programme, product, or intervention such as healthcare professionals or organizational leaders; (ii) subject matter experts, people with specialized knowledge relevant to the outcome; (iii) politicians or decision-makers, those who hold a relevant political role for a given intervention or programme; and (iv) designers, professionals who facilitate the design process of an intervention or programme.[6]

When planning co-design workshops, the recommended number of participants typically ranges from 5 to 12, depending on the workshop's purpose and level of interactivity. However, group dynamics and how participant numbers influence the process are crucial considerations.[31] Rather than prescribing a fixed number, co-design models emphasize the importance of representativeness and diversity to ensure a broad range of experiences.[11,22] In practice, diversity of perspectives often takes precedence over sample size. Groups should be small enough to foster meaningful interaction, yet large enough to reflect varied viewpoints. Decisions about participant numbers are often guided by the concept of information power, which suggests that fewer participants may be needed if their contributions are rich and highly relevant.[23] Co-design studies frequently involve vulnerable populations, making it essential to balance the need for comprehensive input with minimizing participant burden.[24] Ultimately, maintaining a balance between group size and diversity is key to gaining deeper insights.[25] Understanding participants' cultural backgrounds is also vital, as cultural factors can uniquely shape the co-design process.[26,27] For instance, gender and religious beliefs may influence group dynamics. In addition, promoting equity, diversity, and inclusion is especially important in fields such as cardiovascular research.[28]

How to prepare for co-design research?

Preparation for co-design is crucial for effective and meaningful participation throughout the process.[12] To keep participants engaged and productive, participants need to be motivated, focused, and aware of boundaries.[25] Therefore, participants should have a clear understanding of the objectives and expected outcome if interviews or group discussions are being planned. They should also be clear about their roles within the co-design process, which may be achieved by sharing pre-workshop materials. It is important to consider modality and that people may prefer to have information in different ways. This may be a rationale for having written information, informational videos, and questionnaires or having preliminary meetings with participants.[25,29] We recommend ensuring that participants have clear information on how to prepare for co-design activities that can act as an important step in building rapport and avoid initial stress or irritation. Planning healthy refreshments for longer meetings can provide a way to encourage 'natural breaks' where participants can have a chance for rest, refuelling, reflections, or getting to know each other. We also recommend planning for unexpected situations, such as participants arriving late or subgroups unintentionally forming within the larger group. These issues can affect group dynamics, particularly if healthcare professionals, patients, or informal carers are meeting in the same session.

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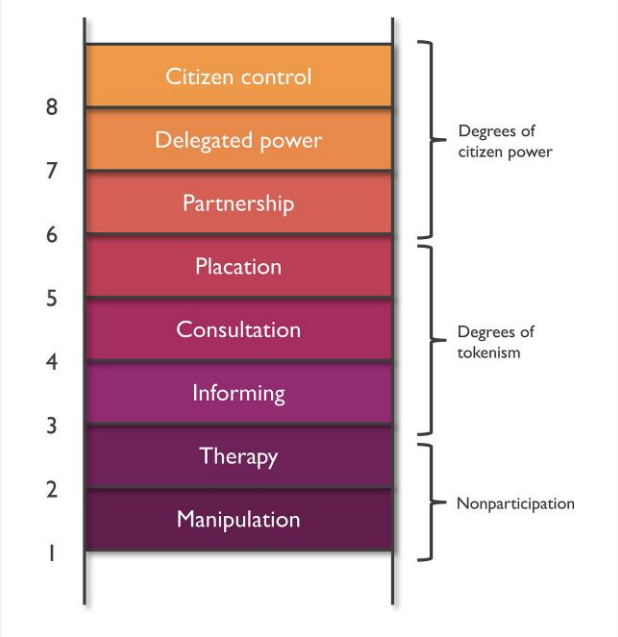


Figure 1 Arnstein's ladder of participation. Adapted from Arnstein, S., 2019. 'A Ladder of Citizen Participation.'[30]

Possible solutions include having predetermined seating to mix groups and a plan for those arriving late. Another aspect of preparing for co-design is to consider the level of participation of end-users and other stakeholders. Different levels of participation refer to how much influence participants will have over what is being co-designed. Arnstein's 'Ladder of Participation' offers a way to think about these varying levels (see *Figure 1*).[30] At the bottom of the ladder, research may manipulate or merely extract data from participants, meaning they are involved in a tokenistic way without any real authority to influence the research. As you move up the ladder, the influence of end-users increases, and at the top, decision-making power is shared with—or even delegated—to the end-users through research activities.[30,32]

From our experience, maintaining a consistent group of co-designers throughout the co-design process can help preserve a clear vision and understanding of the project towards the outcome. That being said, incorporating new perspectives at different stages can be advantageous, as it brings fresh and diverse ideas, helps address emerging challenges, and supports adaptation to the evolving practical needs and dynamics of participants in healthcare.[33] Additionally, allowing some flexibility in group membership can enable individuals in vulnerable positions to join flexibly, as it may help them feel that participation is possible in a way that suits their needs. Nevertheless, it is important that researchers carefully balance various factors when preparing participants for co-design. Without preparation, participants may lack the information needed to engage creatively in the process. However, relying too much on presenting information in, for example, workshops can use up valuable time (which participants seldomly have an abundance of) or overburden those involved if we provide too much information beforehand. Furthermore, not everyone will read the documents or spend as much time preparing as others, which may require that information be repeated during a session that can, in turn, bring frustration into the process.

Pre-meetings can be valuable for preparing co-design participants for a session. Introductory meetings one-to-one or in small groups with the aim of answering any questions the participants may have about the process or topic can be useful.[22,34] Using individual interviews as part of pre-meetings can also capture diverse perspectives and ideas from end-users with different backgrounds. These interviews can ensure that 'quieter' or marginalized voices are not overshadowed in group settings, or if the topic is sensitive, or when group meetings are not possible for any reason. In a co-design study on developing support strategies in patients with cardiac resynchronization therapy, it was found that interviewing patients to brainstorm their perspectives prior to a workshop helped focus on preparation during the workshop.[4] Pre-meetings offer the added advantage of building rapport between participants and co-design facilitators, which can help participants feel more comfortable and engaged during workshops.

Moreover, the research team's role in preparing and moderating sessions is crucial.[5,35] It is essential to foster an environment of mutual respect throughout the co-design process. All participants should be acknowledged as experts in their own right—whether through lived experience, professional knowledge, or contextual insight. Their time, contributions, and perspectives must be genuinely valued.[36] Further, to ensure meaningful engagement, researchers should clearly communicate each participant's role and the purpose of their involvement. Creating space for participants to feel heard and respected is fundamental to the integrity of co-design.[5,35]

We experienced that it is beneficial to give a brief and goal-oriented presentation to participants at the beginning of a co-design session. This can provide participants with the key knowledge required for the workshop, the expectations for the session, and to update co-designers on actions taken since previous sessions. In the preparation, it is also important to discuss who will moderate the session and how to do this.

How to deal with power dynamics?

Managing power dynamics is essential to create a balanced and inclusive environment where input of all participants are heard and valued.[14] When identifying eligible participants for co-design, it is important to be aware of inherent imbalances between end-users and the associated impact on the design, delivery, or evaluation process of the developed intervention. For example, a physician may have more (perceived) power than a patient or another healthcare professional, which may give physicians more input compared with others, or a researcher may hold on to knowledge of data collection and analysis that gives them power over the interpretation and dissemination of the findings. Managing power dynamics in a co-design session is a constant challenge that requires ongoing attention and monitoring from the research team, and we have experimented with different strategies to address these challenges. For example, we have found that organizing separate sessions for patients and professionals could be helpful. Separate sessions allow patients to share ideas more freely, without the influence of potentially dominant professional voices or interference with the relationship between patients and their healthcare professional. Insights from patient meetings can then serve as a basis for professionals' discussions[37] or, of course, vice versa. If the aim of the study will be better addressed using mixed groups, all participants should be informed prior to their participation about the presence of others, including their equal role in the co-design, and assured that participation does not affect, e.g. their treatment. To strengthen equality, it may be useful to have the before-mentioned pre-meetings.[4] Finally, one very practical suggestion is to ask participating healthcare professionals to take off their uniform prior to a meeting to enhance equality among participants.

What to consider about setting?

The physical setting is crucial in the co-design process to help everyone feel comfortable sharing their opinions and foster collaboration, providing spaces and tools that facilitate interaction and creativity.[38,39] Our recommendation is to meet participants in settings that participants can feel comfortable in (e.g. healthcare clinics—where they might feel a sense of familiarity—youth centres, churches, community centres,

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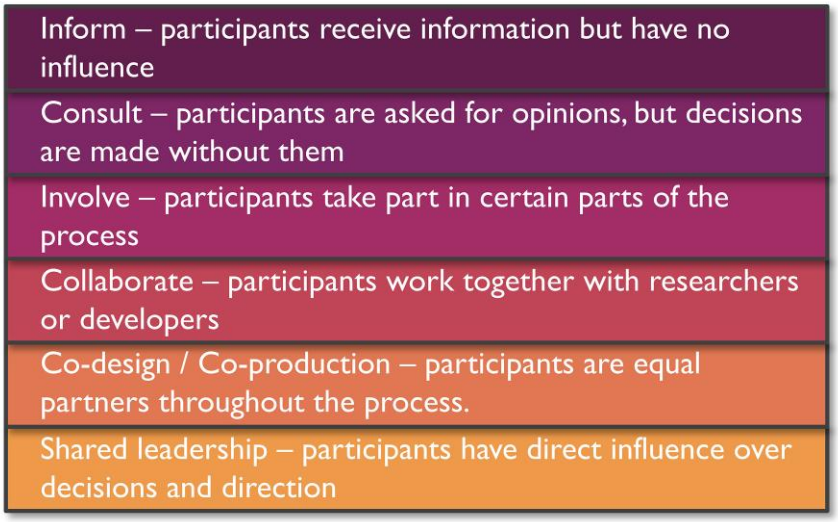


Figure 2 Levels of engagement continuum often used in health research.

or public spaces if feasible). Meeting participants in their own contexts may reduce the burdens involve in participation, and this flexibility can increases the willingness to participate.[24] It can be helpful to offer different ways for the patients to participate to facilitate engagement, for example, coming to their homes, the clinics physically or another place they preferred, or digitally join by Zoom or by phone. In our research, participants sometimes chose digital meeting or telephone due to it being the most practical for them, but sometimes they only wanted to participate if the meetings were in person.

How to keep engagement?

To keep engagement throughout the co-design process is of most importance. However, engagement is not a binary issue (involved or not), but rather exists on a spectrum, ranging from very low to very high. Understanding and using an engagement continuum helps researchers and project leaders to reflect on how much influence participants have, identify opportunities to increase engagement, and avoid tokenism (superficial involvement without real influence). Levels of Patient and Researcher Engagement in Health Research[40] outlines a continuum of six stages of engagement and highlights the equal importance of both patient and researcher involvement throughout the research process. In *Figure 2* you can see common levels of patient and researcher engagement in health research, which often are used as an engagement continuum. Furthermore, tips to enhance engagement are presented in Box 1.

Reimbursement and recognition

In addition, the issue of renumeration for research participants is debated in medical research, with studies examining its ethical, practical, and regulatory dimensions.[41–44] Payment practices vary internationally. In Sweden, participants are usually reimbursed for travel and sometimes lost income, typically after study completion.[44] In the USA, the Institutional Review Board (IRB) ensures payments are not coercive and must approve compensation.[45] Similarly, the UK's Health Research Authority mandates that payments reflect participants' time and effort without being unduly influential.[46] Remuneration can be a way to show respect for participants' time and effort, and could also aid in recruitment and retention, and ensure equity while avoiding undue inducement.[47,48]

Thus, there is a lot to consider in a co-design process, and in *Table 2* we have summarized and present recommendations for methodological considerations when using co-design.

Box 1 Tips to keep engagement throughout the co-design process.

Tips to enhance engagement throughout the co-design process

Preparation of participants

This section emphasizes the importance of getting participants ready before the session begins.

Deepening discussions:

Ensuring participants are informed and comfortable enough to contribute meaningfully.

Tailoring the approach:

Adapting facilitation methods to suit the group's dynamics and individual needs—avoiding information overload or disengagement.

Goal: build confidence and clarity so participants feel safe and motivated to engage.

Planning practicalities for inclusion

This part focuses on the logistical and environmental aspects that influence engagement and includes:

Creating an inclusive atmosphere:

Thoughtful planning of the session's structure and tone.

Practical tools:

Use of name badges, accessible venues or online platforms, seating arrangements, and other inclusive practices.

Goal: remove barriers to participation and foster a creative welcoming space.

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Table 2 Recommendations for methodological considerations in co-design process

Who can be co-designer?	(1) Participants in a co-design process should include those who have a direct interest in the topic and outcome being studied, e.g. patients, healthcare providers, and informal caregivers. One should think broadly and consider also organizations, politicians, or designers as potential end-users. (2) Ensuring diversity perspectives is more important than having many participants. (3) It is valuable to maintain a strong core group of motivated co-designers throughout the process; however, this should not be a closed group. Bringing in fresh ideas can be valuable, and membership of a group should be flexible and responsive to the needs and dynamics of the group.
How to prepare for a co-design process.	(1) Ensure preparation of participants helps to deepen discussions and promote engagement. Tailor the approach based on the group's needs and individual participants to make sure that you do not overwhelm anyone. (2) Seek brevity (one to two pages), accessibility, and readability when using written materials. (3) Offer preparatory materials through alternative formats mediums, i.e. videos, infographics, and animations. (4) Offer introductory pre-meetings to participants to build rapport and to answer questions participants may have. (5) Prepare a brief, goal-oriented presentation for the beginning of each workshop to provide key information for the session. (6) Take time to consider and plan for the practicalities of the session by aiming to create an atmosphere of inclusion that will foster engagement and creativity (e.g. name badges, physical locations vs. online meeting spaces, accessibility arrangements, or seating plans). (7) Consider organizing drinks and refreshments. Providing energy eating or drinking could also provide valuable pauses in discussions. (8) Consider power imbalances in the co-design group before you carry out a co-design activity. This can help you prepare for and attend to the impact of power dynamics in the design process. (9) Consider the co-design setting by focusing on comfort, accessibility, and the facilitation of creative group discussions.
How to conduct co-design.	(1) Tailor the approach based on the group's needs and individual circumstances. This can even be done in partnership with the co-designers by asking them how they think the process should work. (2) Be attentive to power imbalances and plan for strategies on how to support participant involvement.

Completing the co-design and communicating outcomes

Expressing gratitude to participants and communicating the outcomes of a co-design process is an essential ethical practice. It not only recognizes their contributions but also fosters trust and encourages ongoing involvement.[24] Therefore, once the process concludes, it is important to formally acknowledge participants and share the results along with any planned next steps. This gesture can also mark the official end of the co-design initiative. This can be done, for example, through an informational letter or a (an online) meeting. In some cases, participants may also be asked to conduct a final review or to provide their approval of, or input on, the final product. We would also like to encourage researchers to show their appreciation in the acknowledgements section of their publications. Another important aspect of conducting co-design is considering how to communicate findings to the public when feasible. We therefore recommend a communication plan that includes TV, radio, local, and national newspapers, as well as other media and social media.

Conclusion

Co-design is increasingly applied in cardiovascular health research to improve quality and relevance by integrating end-user insights. It supports tailored solutions, empowerment, and self-efficacy while promoting inclusion and collaboration. Despite its benefits, co-design poses practical and methodological challenges, including unclear definitions and evaluation methods. This paper clarifies co-design principles and offers practical tools, examples, and lessons learned. Key considerations include preparing participants, selecting appropriate settings, and addressing power dynamics. Ultimately, co-design occurs when communication and methods enable participants to genuinely influence research outcomes.

Author contributions

Sara Pettersson (Methodology [lead], Writing—original draft [lead], Writing—review & editing [lead]), Hanna Allemann (Methodology [equal], Writing—original draft [equal], Writing—review & editing [equal]), Thomas Grice-Jackson (Methodology [equal], Writing—original draft [equal], Writing—review & editing [equal]), Jane Holstein (Methodology [equal], Writing—original draft [equal], Writing—review & editing [equal]), Heleen Westland (Methodology [equal], Writing—original draft [equal], Writing—review & editing [equal]), and Tiny Jaarsma (Methodology [equal], Writing—original draft [equal], Writing—review & editing [equal])

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